

Causality - SoSe 2025

Introduction

Alexander Marx

Correlation and Causality

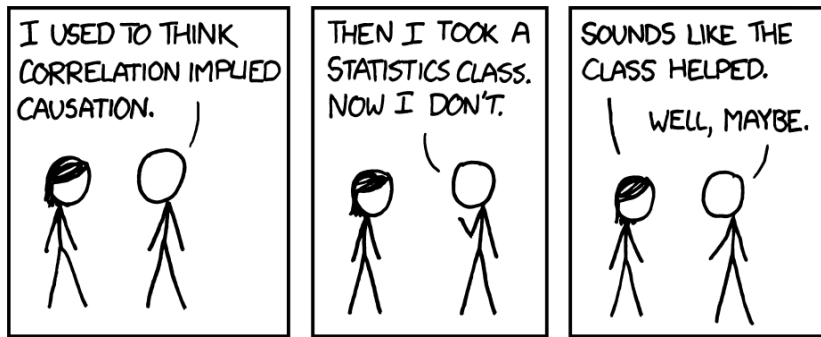
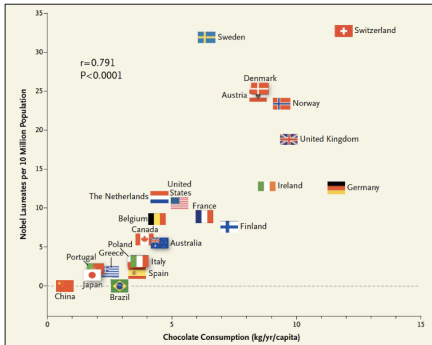
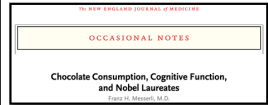
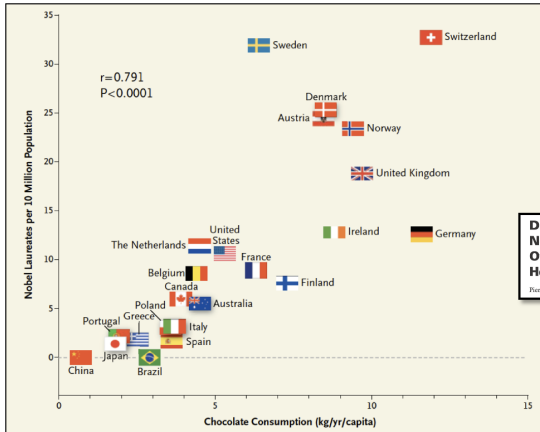


Figure 1: xkcd 552, correlation.png, CC BY-NC 2.5

Correlation and Causality



Correlation does not imply Causality!



Does Chocolate Consumption Really Boost Nobel Award Chances? The Peril of Over-Interpreting Correlations in Health Studies^{1,2}

Pierre Masurel,^{1,2} Alexandre Hoerens,³ and Mauro Pesenti^{1,2,*}

Other Examples

More examples that are rather due to **confounding** (an unobserved common cause), or **selection bias** that affects the data:¹

- The more firemen are sent to a fire, the more damage is done
- Children who get tutored get worse grades than children who do not get tutored
- In the early elementary school years, astrological sign is correlated with IQ, but this correlation weakens with age and disappears by adulthood
- Umbrella sales cause rain

¹Source: from a post of Peter Flom on
<http://stats.stackexchange.com/questions/36/examples-for-teaching-correlation-does-not-mean-causation>

Admin

Admin (1)

Lectures Alexander Marx
Thu 14:15–16:00, CDI/ZHB Room 120

Format Special Lecture, 2V + 1Ü, 4.5 ETCS

Slides On Moodle before every lecture

Communication Please use the Moodle forum to discuss,
ask questions, and form groups

Admin (2)

Tutorials Marlies Hafer

Fri 12:00–14:00, M/E 27

Only 4 dates (**see Moodle for schedule**)

Exercise Sheets Released on Tue (see Moodle)

Hand-in by Tue 23:59 in the following week (via Moodle)

You can hand-in in **groups of size 2**.

Exams 23.07.25 & 24.09.25, 100 minutes, no extra material

Admission Criteria Reach **50% on ≥ 2** out of the first 3 **exercise sheets**

Schedule

Calendar Week	Lecture	Tutorial
15 (07.04.-13.04.)	L1	-
16 (14.04.-20.04.)	L2	-
17 (21.04.-27.04.)	L3	Release of Exercise Sheet 1
18 (28.04.-04.05.)	Public Holiday	T1
19 (05.05.-11.05.)	L4	-
20 (12.05.-18.05.)	L5	-
21 (19.05.-25.05.)	L6	Release of Exercise Sheet 2
22 (26.05.-01.06.)	Public Holiday	T2
23 (02.06.-08.06.)	L7	-
24 (09.06.-15.06.)	L8	Release of Exercise Sheet 3
25 (16.06.-22.06.)	Public Holiday	T3
26 (23.06.-29.06.)	L9	-
27 (30.06.-06.07.)	L10	-
28 (07.07.-13.07.)	L11	Release of Exercise Sheet 4
29 (14.07.-20.07.)	L12	T4

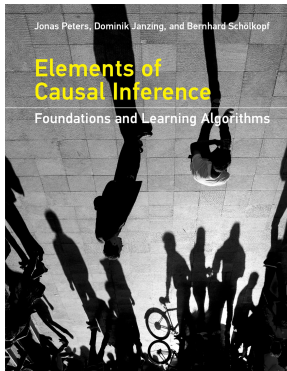
Questions?

This lecture *builds upon and extends* the material from “*Causality*” (WiSe 2022/23) that was kindly provided by **Stefan Harmeling**.

Literature

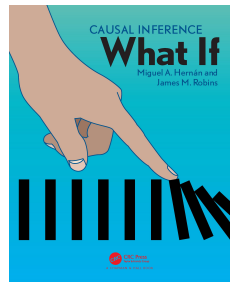
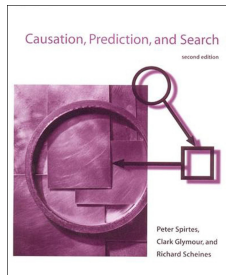
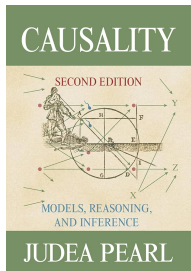
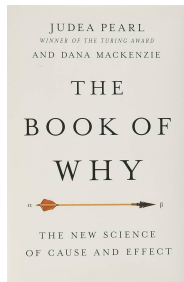
Main Source

The main resource for this course is the following book, for which the PDF is available as open access:



For more details on recent works, individual papers will be referenced occasionally in the corresponding lecture.

Other Great Books



Flow Chart for Causality Books

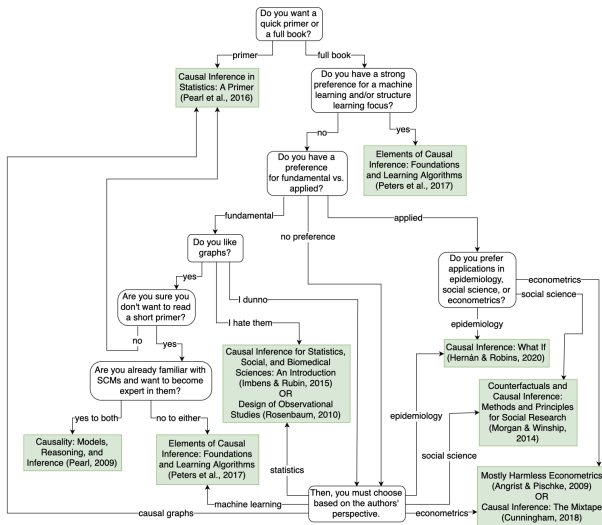


Figure 2: Flow chart by Brady Neal (<https://www.bradyneal.com/which-causal-inference-book>)

Back to Correlation and Causality

Intriguingly high Correlations

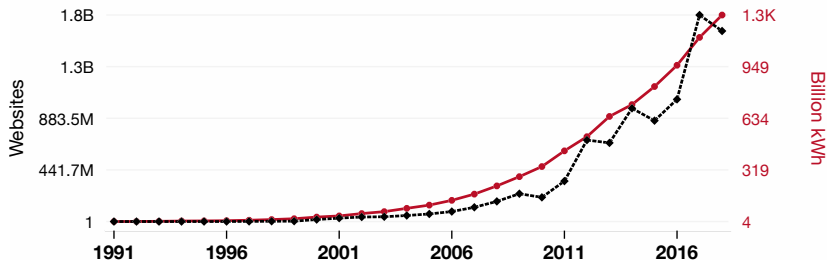
<https://www.tylervigen.com/spurious-correlations>

Intriguingly high Correlations

Number of websites on the internet

correlates with

Total wind power generated globally



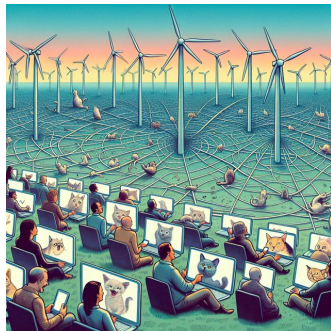
◆--- Number of websites, measured by unique hostnames · Source: Internet Live Stats

●— Total wind power generated globally in billion kWh · Source: Energy Information Administration

1991-2018, $r=0.978$, $r^2=0.957$, $p<0.01$ · tylervigen.com/spurious/correlation/5236

GenAI's Explanation

As the internet expanded, it created a surge in cat video consumption. This led to an increase in laughter around the world, which in turn resulted in stronger and more frequent gusts of wind, ultimately boosting the total wind power generated globally. Remember, laughter is truly the best renewable energy source!



The (Pearson) **correlation coefficient** between two continuous random variables X, Y is defined as

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma(X)\sigma(Y)}$$

where $\text{Cov}(X, Y)$ is the covariance between variables X and Y , and $\sigma(X)$ is the standard deviation of X .

Recall that, the **covariance** is symmetric, i.e.,

$$\text{Cov}(X, Y) = \text{Cov}(Y, X) = E[XY] - E[X]E[Y],$$

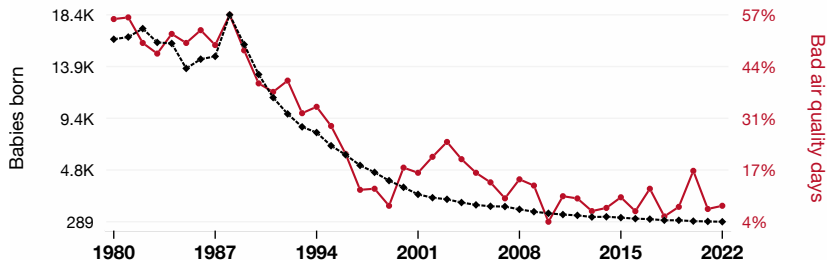
whereas in **causality**, we are often interested in **asymmetries**.

Intriguingly high Correlations

Popularity of the first name Tiffany

correlates with

Air pollution in Los Angeles



◆--- Babies of all sexes born in the US named Tiffany · Source: US Social Security Administration

●— Percentage of days 'unhealthy' or worse air quality in Los Angeles-Long Beach-Anaheim, CA · Source: Environmental Protection Agency

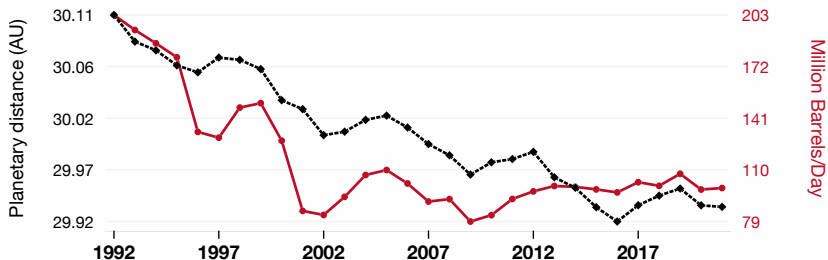
1980-2022, $r=0.966$, $r^2=0.932$, $p<0.01$ · tylervigen.com/spurious/correlation/2183

Intriguingly high Correlations

The distance between Neptune and Mercury

correlates with

Petroluem consumption in Azerbaijan



◆--- The average distance between Neptune and Mercury as measured on the first day of each month · Source: Cacculated using Astropy

●— Volume of petroluem consumption consumed in Azerbaijan in millions of barrels per day · Source: Energy Information Administration

1992-2021, $r=0.798$, $r^2=0.636$, $p<0.01$ · tylervigen.com/spurious/correlation/1058

GenAI's Explanation

As the gravitational pull increased, it created a ripple effect on Earth's magnetic field, leading to improved fuel combustion efficiency in Azerbaijani vehicles.



More on Correlation

Correlation measures only similarity between curves:

- Empirical correlation coefficient ρ is basically a normalized inner product between two vectors
- ρ is a second-order statistics
- ρ views scatter plots through the Gaussian glasses

However, **correlation does not imply causation!**

Nonetheless, correlation *gives hints* to interesting relationships that *can be investigated further*.

Reichenbach's Common Cause Principle (1949)

If two random variables X and Y are **statistically dependent**, then this dependence is either due to a *common cause* Z , or X causes Y , or Y causes X .

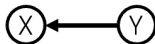
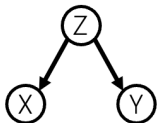


Figure 3: Hans Reichenbach (1891-1953)

Nowadays, we add the following possibilities:

- A **mediator**, e.g., $X \rightarrow Z \rightarrow Y$, or $Y \rightarrow Z \rightarrow X$,
- **Selection bias** affecting both variables X and Y ,
- Or a combination of the above.

But does Causation imply Correlation?

More General Dependencies

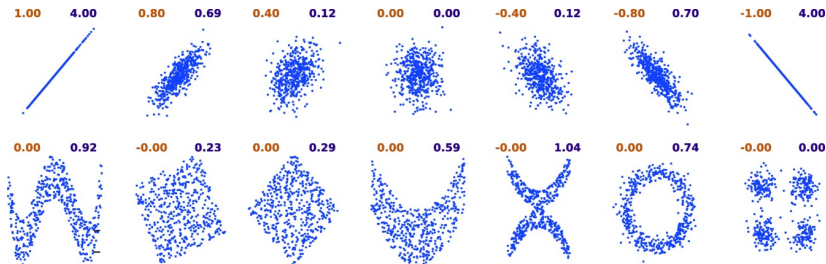


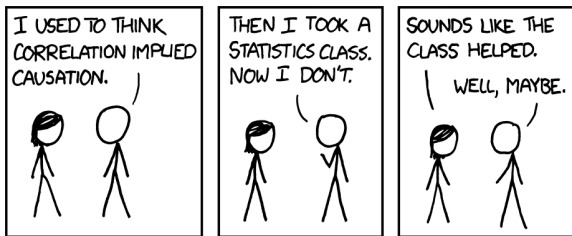
Figure 4: Sets of (x, y) pairs, for which the correlation (left) and the mutual information (right) have been estimated (Ince et al., 2017, DOI: <https://doi.org/10.1002/hbm.23471>).

Pearson correlation captures **linear** dependencies:

- Thus, uncorrelatedness ($\rho = 0$) **does not imply independence**
- Even worse, X and Y could be **uncorrelated, but X causes Y** , e.g. example at the bottom left with X horiz and Y vert

Remarks on Correlation

- We saw that *correlation does not imply causation*, and that the opposite is also not always true
- No correlation does not imply no dependence
- X and Y could be uncorrelated, even though X causes Y
- During the class, we will even see examples in which *causation does not imply dependence* (which is even stronger)



What is the general problem here?

The data and our assumptions!

We try to answer **causal questions** only **based on observed data**.

Pearl's Causal Hierarchy

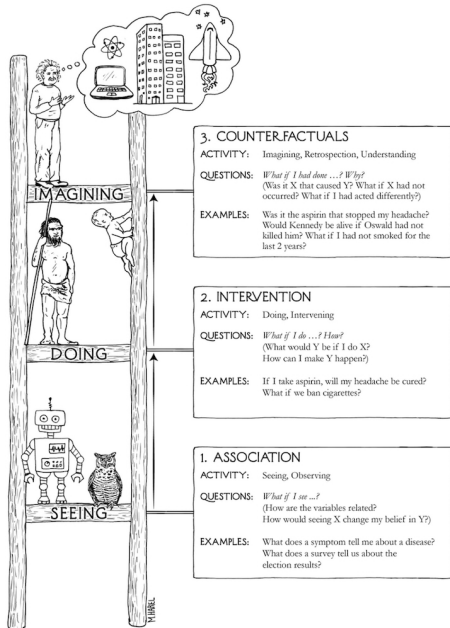


Figure 5: Ladder of Causation, J. Pearl, 2018

$$p(y_{X=x} \mid Y = y', X = x')$$

Requires the causal graph, and the structural causal model.

$$p(y \mid do(X = x))$$

Requires interventional data, or knowledge of the causal graph.

$$p(y \mid X = x)$$

No causal requirements.

Causal Language

Graphs, Functions, and Probabilities

Towards Causal Models: Mechanisms

Studies show that married people are happier.

- Does it mean, marrying makes us happier?
- Or does no one marry unhappy people?

Let's define

M = "being married"

H = "being happy"

Four Scenarios as Words

- If you get married, you are likely more happy:

M causes H .

- If you are more happy, you are more likely to marry:

H causes M .

- There is a common cause C that makes you more likely to be happy and to get married:

C causes H and M .

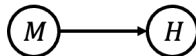
- Being married and being happy has nothing to do with each other:

H and M are independent.

Four Scenarios as Graphs

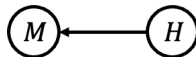
- If you get married, you are likely more happy:

$$M \rightarrow H$$



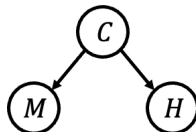
- If you are more happy, you are more likely to marry:

$$H \rightarrow M$$



- There is a common cause C that makes you more likely to be happy and to get married:

$$M \leftarrow C \rightarrow H$$



- Being married and being happy has nothing to do with each other:

$$M, H$$



Four Scenarios as Computer Programs

- If you get married, you are likely more happy:

```
M = f(); H = g(M);
```

- If you are more happy, you are more likely to marry:

```
H = f(); M = g(H);
```

- There is a common cause C that makes you more likely to be happy and to get married: $C = f(); M = g(C); H = h(C);$

- Being married and being happy has nothing to do with each other:

```
M = f(); H = g();
```

Four Scenarios as Probabilities

- If you get married, you are likely more happy:

$$P(M, H) = P(M) P(H|M)$$

- If you are more happy, you are more likely to marry:

$$P(M, H) = P(H)P(M|H)$$

- There is a common cause C that makes you more likely to be happy and to get married:

$$P(M, H) = \sum_{c \in \mathcal{C}} P(C = c) P(M|C = c) P(H|C = c)$$

- Being married and being happy has nothing to do with each other:

$$P(M, H) = P(M)P(H)$$

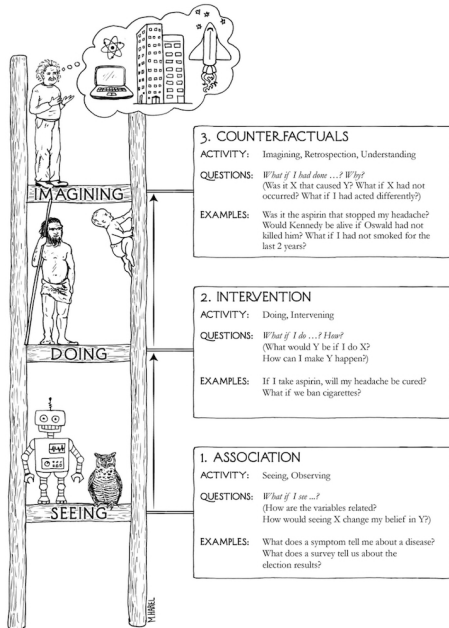


Figure 6: Ladder of Causation, J. Pearl, 2018

$$p(y_{X=x} \mid Y = y', X = x')$$

Requires the causal graph, and the structural causal model.

$$p(y \mid \text{do}(X = x))$$

Requires interventional data, or knowledge of the causal graph.

$$p(y \mid X = x)$$

No causal requirements.

Observational Studies

Example:

- Smoking is associated with diseases
- But does it **cause** diseases?
- We cannot force people to smoke (**do an intervention**)
- There are potential confounders like gender, age, . . .

What can we do?

- Compare similar subgroups (e.g., males who smoke to males who don't)
- **Control for confounders**, but how?

Causality and Simpson's Paradox (Observational Data)

Overall	
Treatment <i>a</i> : Open surgery	78% (273/350)
Treatment <i>b</i> : Percutaneous nephrolithotomy	83% (289/350)

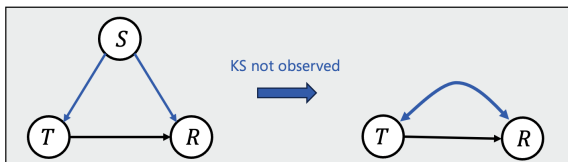
Figure 7: Simpson's paradox: Recovery rate after different treatments for kidney stones

Causality and Simpson's Paradox (Observational Data)

	Overall	Patients with small stones	Patients with large stones
Treatment <i>a</i> : Open surgery	78% (273/350)	93% (81/87)	73% (192/263)
Treatment <i>b</i> : Percutaneous nephrolithotomy	83% (289/350)	87% (234/270)	69% (55/80)

Figure 8: Simpson's paradox: Recovery rate after different treatments for kidney stones

Causal view: (Unobserved) **confounding** through kidney stone size:

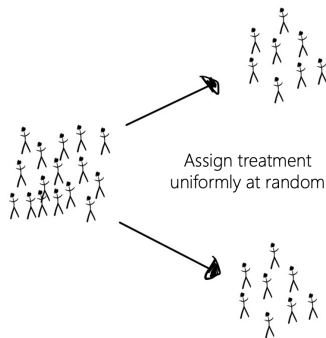


Randomized Controlled Trials

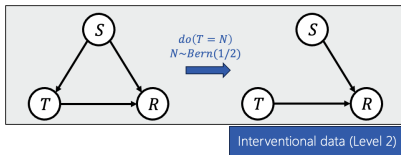
Gold standard for causal inference:

- A study that includes treated individuals and **controls**
- **Controlled experiment** and clear experimental design
- Treatment assignment is **randomized**
- The study is **double blind**

Randomized Controlled Trials (Interventional Data)



RCT as an **intervention** on the data generative process

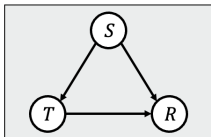


Such data allows for computing, e.g., **average treatment effects**:

$$\begin{aligned} ATE &= E[R \mid do(T = a)] - E[R \mid do(T = b)] \\ &\stackrel{\text{RCT}}{=} E[R \mid T = a] - E[R \mid T = b] \end{aligned}$$

What if we cannot do a RCT?

If we **know the causal graph**, we can check, e.g., via the **backdoor** or **frontdoor criterion** whether our causal query is **identifiable** from observed data, etc. In our case, if we have:



&

	Overall	Patients with small stones	Patients with large stones
Treatment a: Open surgery	78% (273/350)	93% (81/87)	73% (192/263)
Treatment b: Percutaneous nephrolithotomy	83% (289/350)	87% (234/270)	69% (55/80)

We can identify $P(R \mid do(t))$ via the backdoor adjustment, i.e.,

$$P(R \mid do(t)) = \sum_s P(R \mid T = t, S = s)P(S = s)$$

$$P(R = 1 \mid do(T = a)) \approx 0.93 \frac{357}{700} + 0.73 \frac{343}{700} = 0.832, \quad P(R = 1 \mid do(T = b)) \approx 0.782$$

We will discuss this in much more detail in the following lectures.

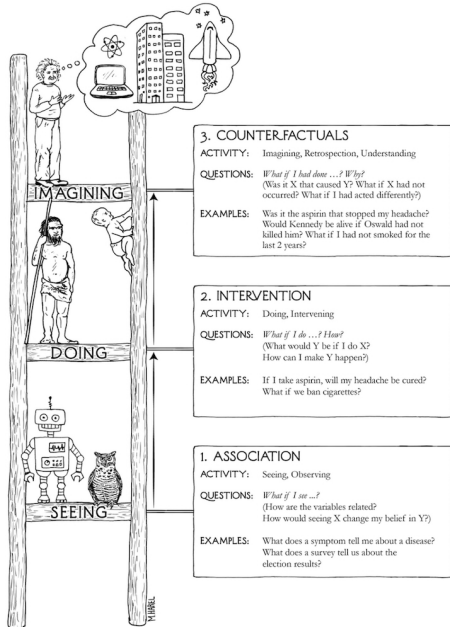


Figure 9: Ladder of Causation, J. Pearl, 2018

$$p(y_{X=x} \mid Y = y', X = x')$$

Requires the causal graph, and the structural causal model.

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Causal Discovery

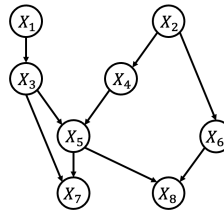
Can learn causal graphs from data?

Given Observational Data

Causal Graph

X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
-1.164294838	0.1315158938	1.58823116792	2.256584649	2.344482611	-0.8508841577	0.187766884	-0.189548880
-1.395184841	0.8713703728	2.0417832952	0.646994844	1.94877622	-0.1808971487	0.7548158889	1.712846466
0.669429152	0.5834298314	-0.2833515157	0.266877236	1.281595079	0.1495888881	1.887956672	-0.415981548
0.281769152	-0.127366437	-1.7297251688	1.538983455	0.91579197	-0.4805514485	2.814992428	0.25541212
0.856998838	-1.184866493	-0.3318413158	-0.297364895	-0.88482615	0.8683851315	-0.84919182	0.209348383
-0.22823832	-1.2234724951	0.2872819233	-0.431887748	-0.3277688	-0.2215134888	-0.854887281	-0.352958855
0.466178978	0.5844627557	0.5897212579	0.571882532	1.43714958	0.8241722327	1.889286148	0.498476879
0.222655156	-0.1236837857	1.695395888	1.730585783	1.68515565	-0.897873183	-0.95451128	-0.602615515
0.386681164	-1.1514173625	2.8688422253	2.256694679	3.1577515	-0.8419517631	-0.843925815	-0.809388462
0.483298786	1.1984513218	1.6882312814	2.861521916	0.45289161	-0.8878714857	0.778899022	-0.817972379
0.410897488	1.7882178893	0.8767878056	0.870997812	1.67461615	0.8996882879	0.886114281	0.337343351
1.475634379	0.7215808886	2.5444725617	4.815127748	0.35365523	-0.8407818866	3.21861271	-0.255484961
0.588127128	-0.284880188	0.791242579	1.351048416	-0.74870984	1.8564614347	-0.258789884	0.738176448
0.764288283	0.4737950882	-0.5194828581	0.234798748	2.83946678	-0.8868848471	-0.88471683	1.168187128
0.543789778	0.878111134	0.8739748328	0.617638884	1.77887795	1.375651278	1.39871718	2.299785886
0.077309151	0.8882830833	0.3848168156	0.131322887	-0.7474658	-0.261482195	-0.83249527	-0.147215867
1.18151746	0.1886677111	1.645148888	1.55921763	1.83588003	-0.115634949	-1.84123821	-0.808146471
1.364784215	-0.4449776574	1.586335127	1.441391457	1.58496114	0.8146897487	-0.944848494	-0.133623852
0.37177871	1.3588868714	0.5613788964	0.913148886	1.148184972	0.8238214647	-0.434512089	0.981793964
0.17814711	0.8775881983	-0.673381866	-0.86054618	1.33525664	0.8584337577	-1.277670803	0.567949378
-0.41642877	-1.5282162177	-0.9194423637	-1.355881541	1.87161499	-0.8213835888	-0.851284114	0.820824956
-1.480463148	-1.318616368	-0.6148816365	-0.89956286	1.788484578	-0.8862971542	-0.83185415	-0.861948487
-1.386234839	-0.8327649718	0.2844181933	-0.521894635	0.48523332	0.8979884978	-1.821860285	-0.839874784
-0.216778779	-0.212456178	0.7881564942	-1.39816572	1.599778888	-1.197751429	-0.287264566	-0.287264566
1.644721146	-0.2888862617	-0.8852821803	1.5880821256	1.78277812	-0.8431614688	-0.35749277	-0.35749277
-0.436376472	-0.2218378982	-0.9772848998	-0.943077627	-0.11289663	-0.1978812972	0.882491828	-0.308845522
-0.187488936	-0.3877861438	-0.7338742567	-0.826485782	-0.88488504	0.8871343846	-0.881748064	-1.173778441
0.818229922	-0.2139548832	0.1588885561	1.817038478	0.85578258	0.8186281487	1.885174728	-1.272234243
-0.360573133	-0.8438878955	0.51382748512	-0.167756178	-0.52181628	-0.6801118871	-1.431762331	1.802524166
-0.494789286	-0.1381758298	-0.1688474887	-0.863356448	2.93733087	-0.8428748757	0.519614373	1.383938263

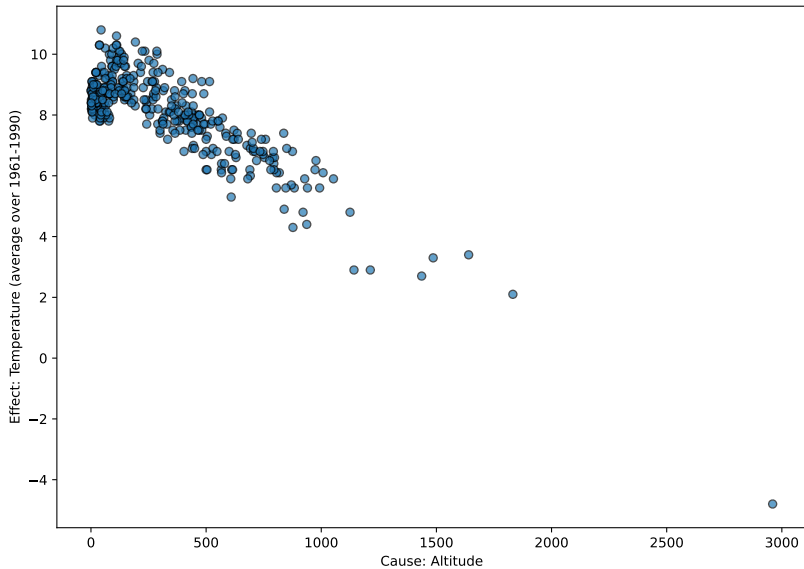
Learn



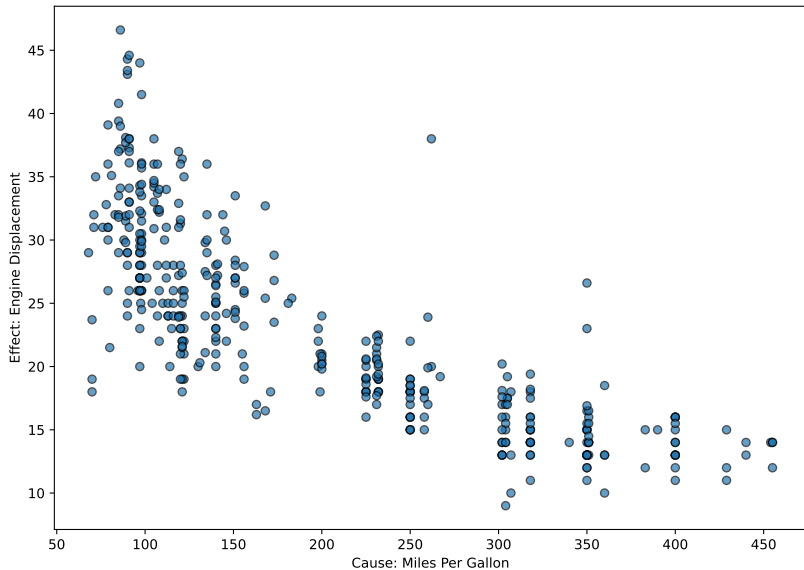
Tübingen Cause-Effect Pair Challenge

Source: Mooij, Peters, Janzing, Zscheischler, Schoelkopf. *Distinguishing cause from effect using observational data: methods and benchmarks*, JMLR, 2016

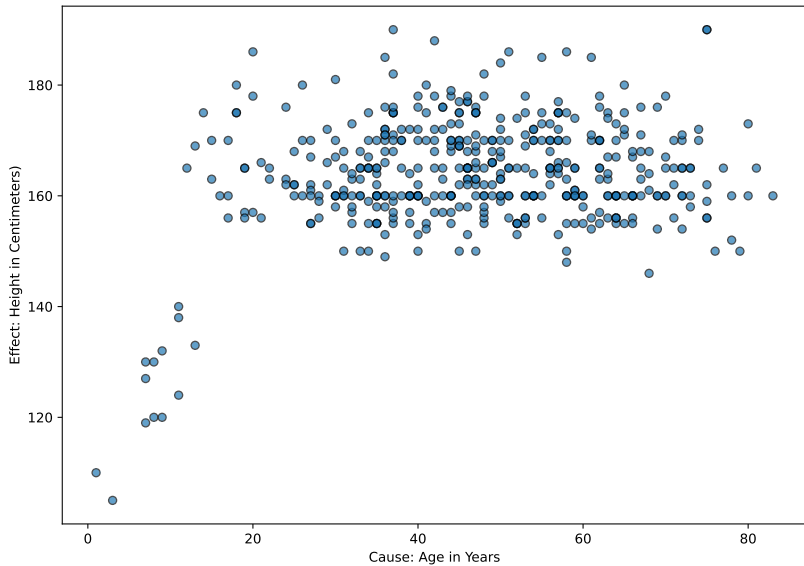
Pair 1



Pair 13



Pair 22



Pearl's causal hierarchy tells us that we **cannot overcome correlation** and infer cause and effect from observed data.

Thus, we need:

- Data from interventions, or
- **Assumptions** that **connect distributional properties with graphical properties**, and/or about the **structural causal model** (e.g., only linear non-Gaussian additive noise functions).

These assumptions should be **verifiable**, **motivated by domain knowledge**, or **hold with high probability**.

Take Home Message

Although statistics began in the 1800s, Sewall Wright was the first person (in 1920) to put down mathematically the assumption that X causes Y and not the other way around (Wright, 1921). Using “path diagrams” he was able to articulate causal assumptions mathematically, communicate them and defend them on scientific grounds. Moreover, given the structure of the diagram and its path coefficients, he could calculate correlations among measured variables. Subsequently he worked backwards and found the coefficients from the correlations.

In retrospect, Wright’s exercise was remarkable (Pearl and Mackenzie, 2018). Everybody jumped on him for doing the impossible, extracting causation from correlation. But he responded admirably, claiming that his causal conclusions emerge not from correlations alone, but from a combination of correlations and causal assumptions. This is the same philosophy that rules causal inference today. Yet even today, many researchers still have a hard time understanding how path diagrams can do things that correlations alone cannot (Sobel, 2009); they haven’t been taught how to read the causal assumptions that are so vividly displayed in the diagram.

Figure 10: Excerpt of an Interview with Judea Pearl (in “Observational Studies”, 2022)

Summary

During the course of this lecture, we will cover the following topics:

- **Graphical models** (Causal Bayesian Networks)
- **Pearl's do-calculus**
- **Causal inference** (covariate adjustments, e.g., backdoor criterion)
- **Causal discovery** (classical algorithms, e.g., PC algorithm, assumption violations, etc.)
- Aspects of **causal machine learning**

Thank you!